		$\phi_M = \frac{0.012}{0.25} \times (-63) = -3\ MJ\ kg^{-1}$
12	**B**	Particle in simple harmonic motion $a_0 = \omega^2 x_0$ $0.30 \times 10^{-2} = \omega^2 \times 6 \times 10^{-2}$ $\omega = 0.22\ rad\ s^{-1}$ or $\omega^2 = \lvert gradient \rvert$
13	**C**	P is at the positive max position at t = 0, so cosine function. Phase difference $= 2\pi \frac{\Delta x}{\lambda} = 2\pi \frac{0.75\lambda}{\lambda} = 1.5$